

DEBAJYOTI MAZUMDER

PH.D. SCHOLAR · NLP

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Summary

Debayyoti Mazumder is a Ph.D. scholar in Data Science and Engineering at IISER Bhopal, working under the guidance of Dr. Akash Anil. His research focuses on code-mixed NLP, a setting within the broader areas of multilingual and low-resource NLP. His Ph.D. focuses on the synergy between code-mixed setting and its constituent native languages. He has published research in transactions such as ACM TALLIP, conferences like EMNLP, WSDM and ACL-IJCNLP, and workshops including DHOW-MiLLA@TheWebConf. His industry experience spans LLM reasoning and reinforcement learning at NielsenIQ and recommender systems at Adobe.

Education

Ph.D. in Data Science & Engineering

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH - BHOPAL (IISERB) | CPI: 9.00

Supervisor: Dr. Akash Anil

Aug. 2022 - present

Madhya Pradesh, India

M.Sc. in Mathematics

ZAKIR HUSSAIN DELHI COLLEGE, UNIVERSITY OF DELHI | 69.25%

Jul. 2018 - Oct. 2020

New Delhi, India

B.Sc. (Honours) in Mathematics

DESHBANDHU COLLEGE, UNIVERSITY OF DELHI | SGPA: 8.70

Jul. 2015 - Jul. 2018

New Delhi, India

Work Experience

Adobe

RESEARCH PHD INTERN

July, 2026 - Sept, 2026

- Next-question recommender for CXOs.

NielsenIQ

RESEARCH SCIENTIST - SR. INTERN

July, 2025 - Jan, 2026

- LLM Reasoning using RLVR techniques.
- Warm up SFT and reasoning on ecommerce-based tasks.

Publications

Entailed Opinion Matters: Improving the Fact-Checking Performance of Language Models by Relying on their Entailment Ability

G. Kumar, A. Garg, D. Mazumder, A. Kishore, B. kumar, J. Patro

FACT CHECKING (PRE-PRINT: [HTTPS://ARXIV.ORG/PDF/2505.15050](https://arxiv.org/pdf/2505.15050))

ACL-IJCNLP Findings 2026

- Leveraging entailment ability of language model.
- Thorough comparison between training and prompting based methods.

To Trust or Not to Trust: Retrieval-Augmented Fact Checking in Speech

D. Mazumder, Mamta, A. Subramanyam

FACT CHECKING IN SPEECH (PRE-PRINT: [HTTPS://ARXIV.ORG/ABS/2609.30227](https://arxiv.org/abs/2609.30227))

EMNLP-2026

- Transferability of of fact verification ability of LALMs from text to speech.

Evaluating Cross-lingual Knowledge Consistency in Code-Mixed vis-a-vis Indian Languages using IndicKLAR

D. Mazumder, D. Pathak, P. Kodali, A. Joshi, A. Agarwal, J. Patro

CROSSLINGUAL CONSISTENCY (PRE-PRINT: [HTTP://ARXIV.ORG/ABS/2605.29637](http://arxiv.org/abs/2605.29637))

EMNLP Findings-2026

- IndiKLAR, an Indic benchmark of covering 18 Indian languages.
- Translate-in-Thought (*TinT*) prompting reduce gap between low-resource and high-resource languages.

Neither Here Nor There: Cross-Lingual Representation Dynamics of Code-Mixed Text in Multilingual Encoders

D. Mazumder, D. Pathak, P. Kodali, J. Patro

CROSSLINGUAL REPRESENTATION (PRE-PRINT: [HTTPS://ARXIV.ORG/PDF/2603.19771](https://arxiv.org/pdf/2603.19771))

EMNLP Findings-2026

- Trilingual corpus enabling probe of cross-lingual representation alignment across code-mixed and native languages.
- Proposed post-training alignment motivated by interpretability analysis.

On VLMs for Diverse Tasks in Multimodal Meme Classification

D. Gavit, G. Kumar, D. Mazumder, S. Das, J. Patro

MULTIMODAL MEME CLASSIFICATION (HTTPS://DL.ACM.ORG/DOI/10.1145/3774905.3796487)

DHOW-MiLLA@TheWebConf 2026

- A comprehensive study of state-of-the-art vision-language models (VLMs).

Mind the Links: Cross-Layer Attention for Link Prediction in Multiplex Networks

D. Sharma, A. Kishore, A. Garg, D. Mazumder, D. Mohapatra, J. Patro

LINK PREDICTION IN MULTIPLEX NETWORKS (HTTPS://DL.ACM.ORG/DOI/10.1145/3773966.3779365)

ACM WSDM - 2026

- Novel task : Link prediction task in partial graph of multiplex networks.

Revealing the impact of synthetic native samples and multi-tasking strategies in Hindi-English code-mixed humour and sarcasm detection

D. Mazumder, A. Kumar, J. Patro

MULTI-TASK LEARNING FRAMEWORK FOR HUMOR & SARCASM DETECTION

EMNLP Findings-2025

(HTTPS://ACLANTHOLOGY.ORG/2025.FINDINGS-EMNLP.1308/)

- Creating a multi-task learning architecture that leverages shared knowledge among different tasks.
- Compared the multi-task learning framework with the current state-of-the-art models.

Can native language samples improve code-mixed hate detection?: A case study for Hindi-English code-mixed hate detection

D. Mazumder, A. Kumar, J. Patro

CODE-MIXED HATE DETECTION (HTTPS://DL.ACM.ORG/DOI/10.1145/3726866)

ACM TALLIP - 2025

- Adding native hate samples in the training set for code-mixed hate detection.
- Transferring knowledge from native hate samples alone to detect code-mixed hate.

Roles & Honors

2026	Student Volunteer and D&I, EMNLP 2026	Budapest, Hungary
2026	ACM/IARCS Travel Grant, EMNLP 2026	Budapest, Hungary
2026	Selected for Machine Learning Summer School, MLSS NYC 2026 (Columbia University)	NY, USA
2026	SIGIR Student Travel Grant, WSDM 2026	Boise, Idaho, USA
2026	ACM Travel Grant (Invited talk and poster presenter), ACM ARCS 2026	IITH, INDIA
2025	ANRF ITS Travel Grant (Poster Presenter), EMNLP 2025	Suzhou, CHINA
2025	Selected for LiveRAG Challenge track, SIGIR 2025	Padua, Italy
2023	Travel Grant (Participant), Indian Symposium on Machine Learning (IndoML – 2023)	IITB, INDIA
2023	2nd Position, Poster Presentation on DSE(Data Science & Engineering) day	IISERB, INDIA
2022	Travel Grant (Participant), Indian Symposium on Machine Learning (IndoML – 2022)	IITGN, INDIA
2022	MHRD Fellowship Recipient, GATE 2022 Exam	INDIA

Reviewer: ICWSM 2026 | The ACM Web Conference 2026, 2027 | LT-EDI-2026 | DravidianLangTech-2026 | IEEE Transactions on Multimedia | INSTCon 2026

Courses

IISER BHOPAL

- Machine Learning- DSE617, Deep Learning- DSE616, Natural Language Processing- DSE607, Advance Natural Language Processing- DSE 618, Spatial Data Science & Applications- DSE606, Data Science in Practice- DSE615
- Teaching Assistant : Natural Language Processing- DSE318/401/607 (Jan,2026 - Apr,2026), Introduction to Programming- ECS102 (Jan,2025 - Apr,2025), Natural Language Processing- DSE407/607 (Aug,2024 - Nov,2024), Introduction to Programming- ECS102 (Jan,2024 - Apr,2024), Natural Language Processing- DSE407/607 (Aug,2023 - Nov,2023), Applied optimization- DSE311 (Jan,2023 - Apr,2023)